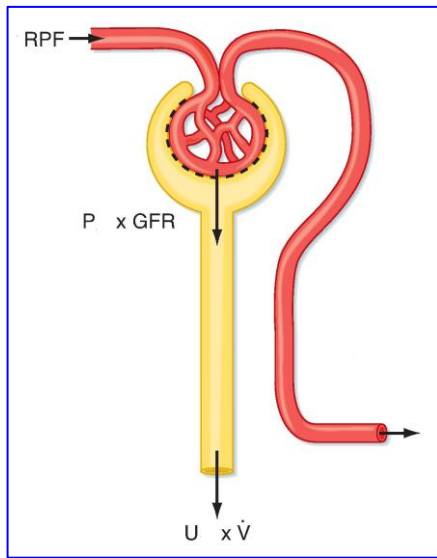
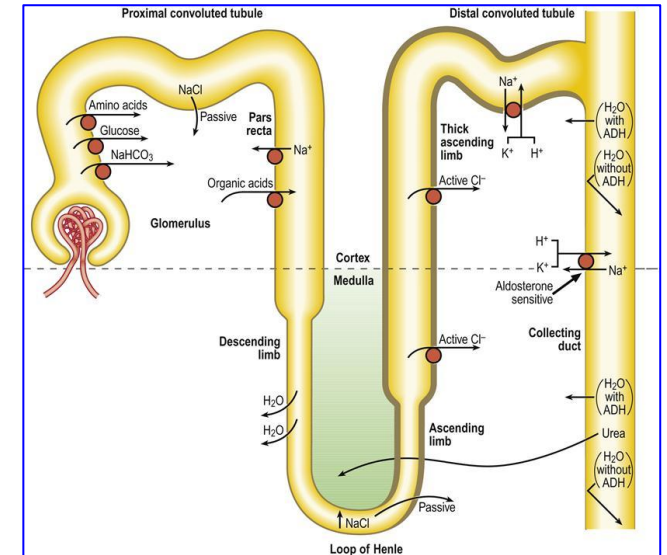




Basic Principles of Medicine 1

Module : Cardio, Pulmonary, Renal

CPR: Lecture 45



Tubular Mechanisms

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0283. Define and understand the concepts of net transport: filtration, reabsorption, secretion, excretion.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0285. Distinguish between active (primary and secondary) transport, facilitated diffusion, and passive diffusion based on energy source and carrier protein involvement.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0286. Explain the contribution of the major nephron segments (PT, LOH, DCT, CT) to the reabsorption of the filtered load of solutes and water.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0288. Describe the function of the following renal transporters and their predominant localization along the tubules with regard to nephron segment and apical versus basolateral membranes: a. Transport ATPases (Na^+/K^+ -ATPase, H^+/K^+ -ATPase, H^+ -ATPase, and Ca^{+2} -ATPase); b. Ion and water channels (K^+ , ENaC, Cl^- , Ca^{+2} , aquaporins); c. Coupled transporters (Na^+ -glucose, Na^+/H^+ -antiporter, $\text{Na}^+/\text{K}^+-2\text{Cl}^-$ -symporter, Na^+ -phosphate symporter, Na^+/Cl^- -symporter, $\text{Na}^+/\text{HCO}_3^-$ -symporter, $\text{Cl}^-/\text{HCO}_3^-$ -antiporter).

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0287. Describe the cellular mechanisms for the transport of Na^+ , Cl^- , K^+ , HCO_3^- , Ca^{+2} , phosphate, organic solutes (e.g., glucose, amino acids, and urea), and water by the major tubular segments.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0289. List the nephron sites (PT, LOH, DCT, CT) and explain the molecular mechanisms of action of the following classes of diuretics (carbonic anhydrase inhibitors, loop, thiazide, K^+ -sparing).

SOM.MK.II.BPM1.3.CPR.2.PHYS.0290. For the major solutes (glucose, amino acids, Na, K, Ca, Mg, H_2O) of the glomerular filtrate and for PAH and inulin, list the areas of the nephron where these molecules are reabsorbed and secreted.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0291. Explain the effects of diuretics on Na^+ handling by the kidneys and, thus, on ECF volume regulation.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0292. Outline the effects (up to five) of hormones on the various segments of the nephron.

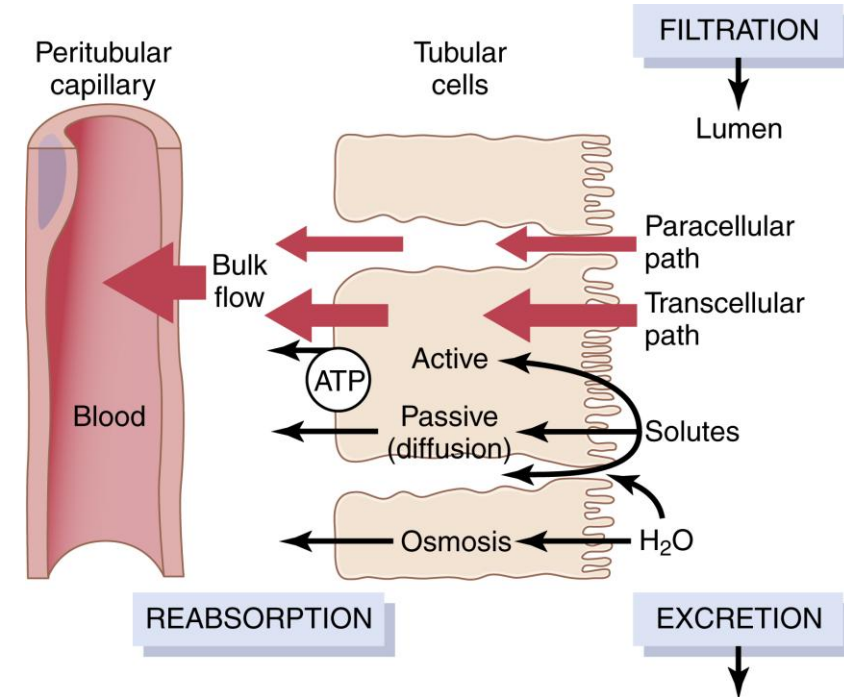
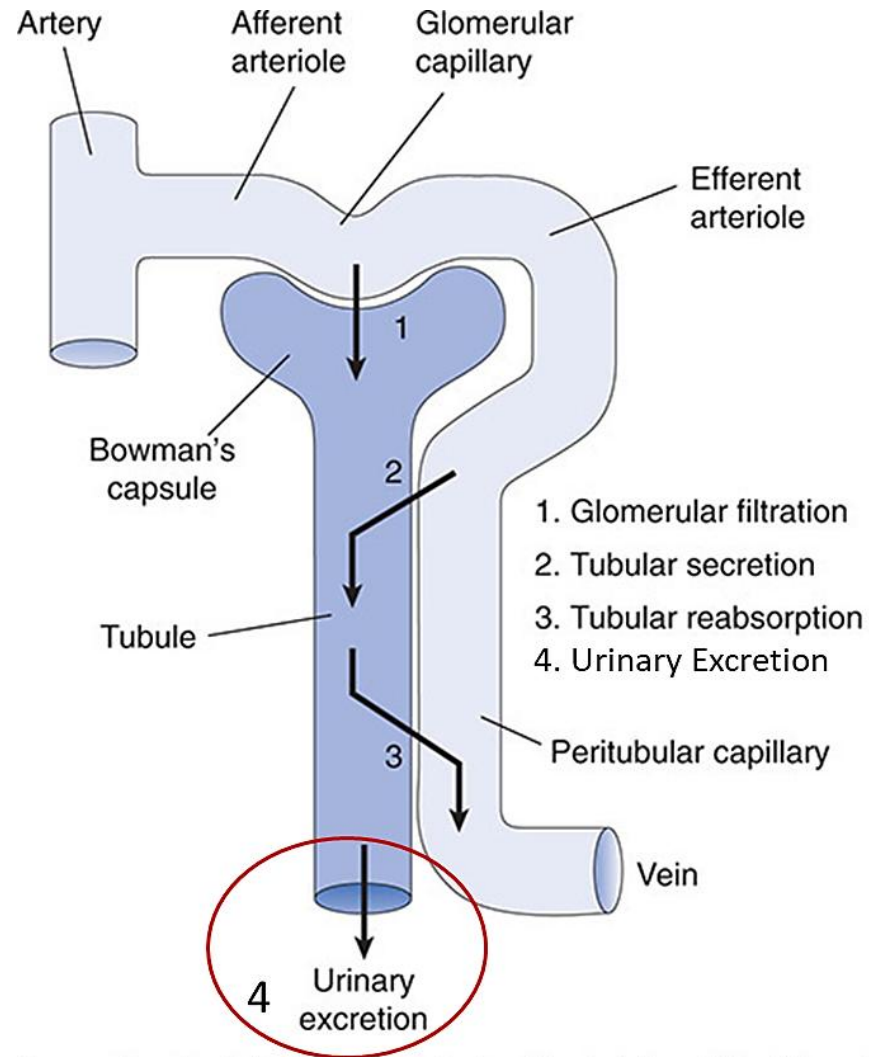
Recommended Readings

Chapter 22: Kidney Functions. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 23: Regulation of Fluid and Electrolyte Balance. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

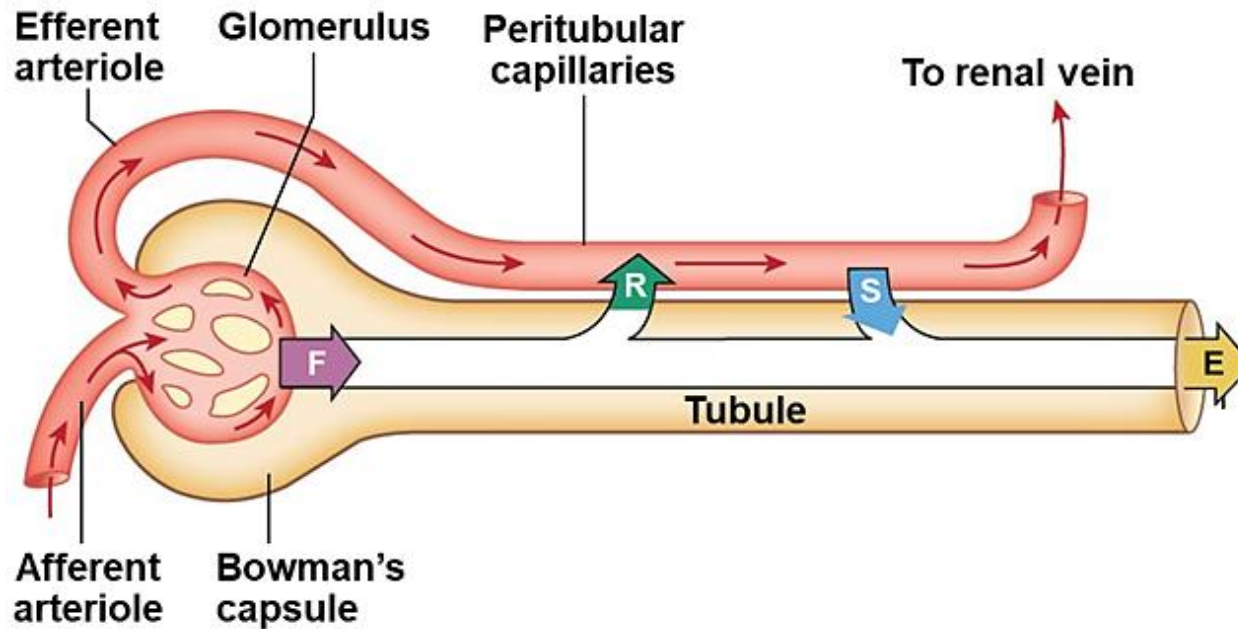
Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Major tubular mechanisms: Reabsorption and Secretion



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology*, 9e
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Quantification of amount filtered and excreted



$$\begin{array}{ccccccc} \text{Amount} & & \text{amount} & & \text{amount} & & \text{amount of solute} \\ \text{filtered} & - & \text{reabsorbed} & + & \text{secreted} & = & \text{excreted} \\ \text{F} & & \text{R} & & \text{S} & & \text{E} \end{array}$$

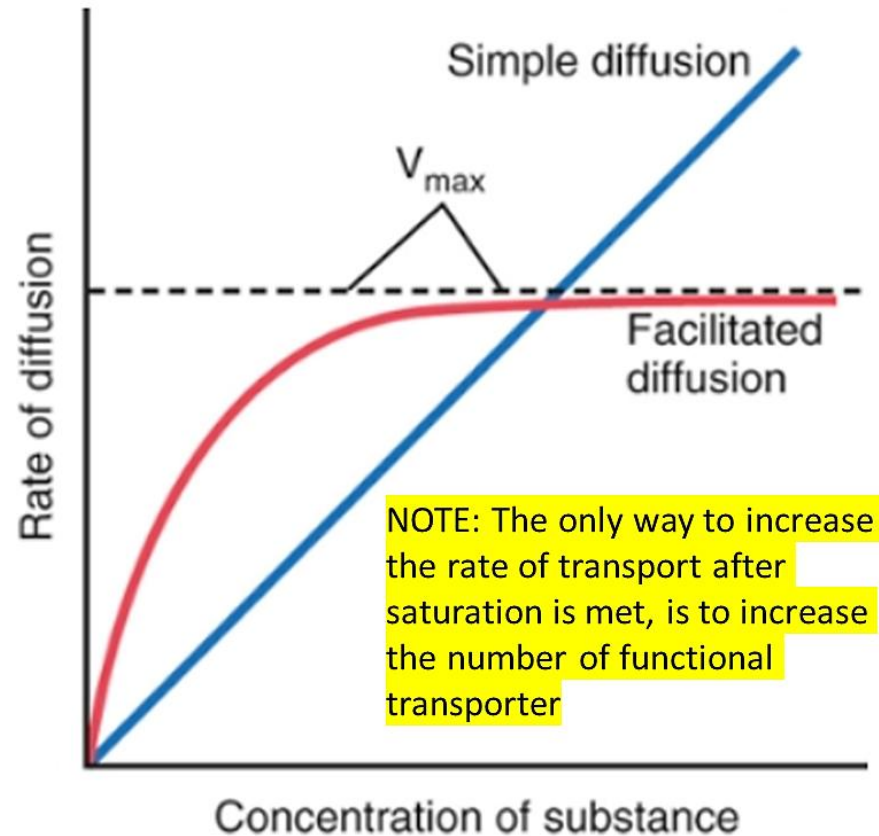
Amount of a substance filtered is called
Filtered Load (FL)
Filtered Load = GFR x Plasma concentration
of the substance

$$\text{FL} = \text{GFR} \times P_{\text{substance}}$$

Amount of a substance excreted in urine is
calculated by,
Amount Excreted = Volume of urine x Urinary
concentration of the substance

$$\text{Amount excreted} = V \times U_{\text{substance}}$$

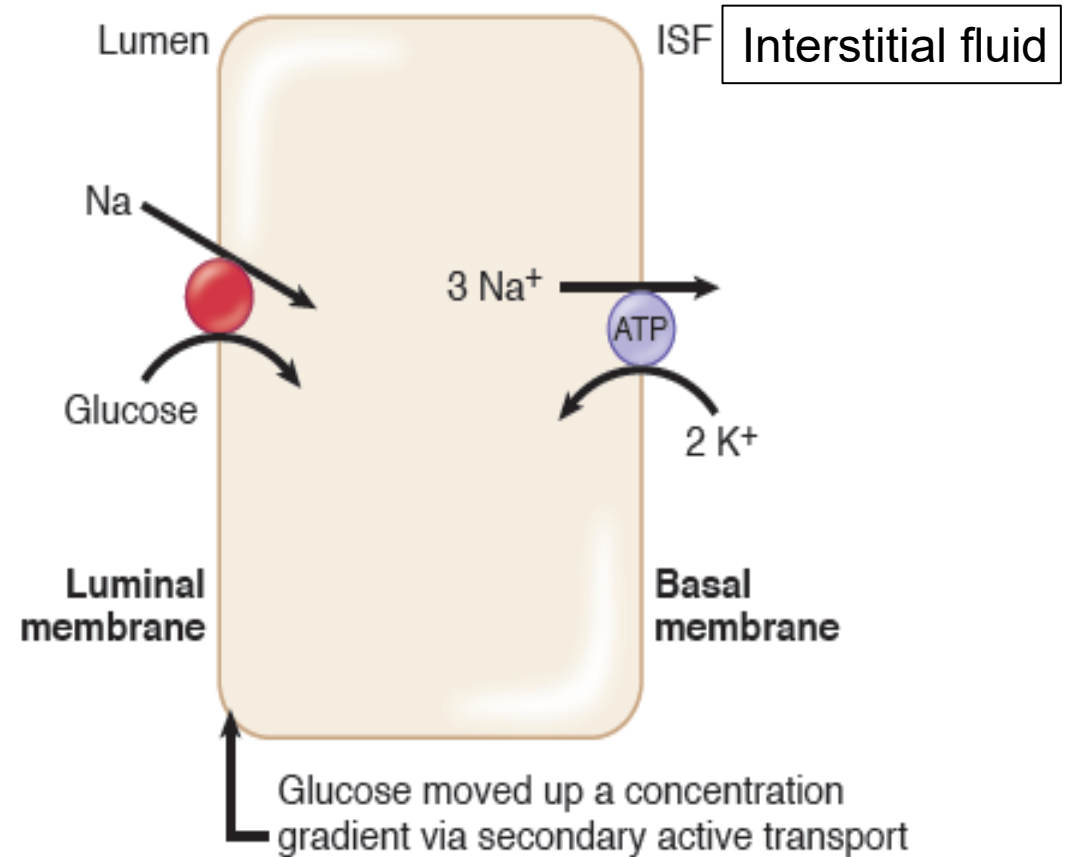
Simple vs Facilitated diffusion



Rate of transport depends on

1. Concentration of solute
2. Number of functioning transporter

Primary vs Secondary Active Transport



Simple vs Facilitated diffusion

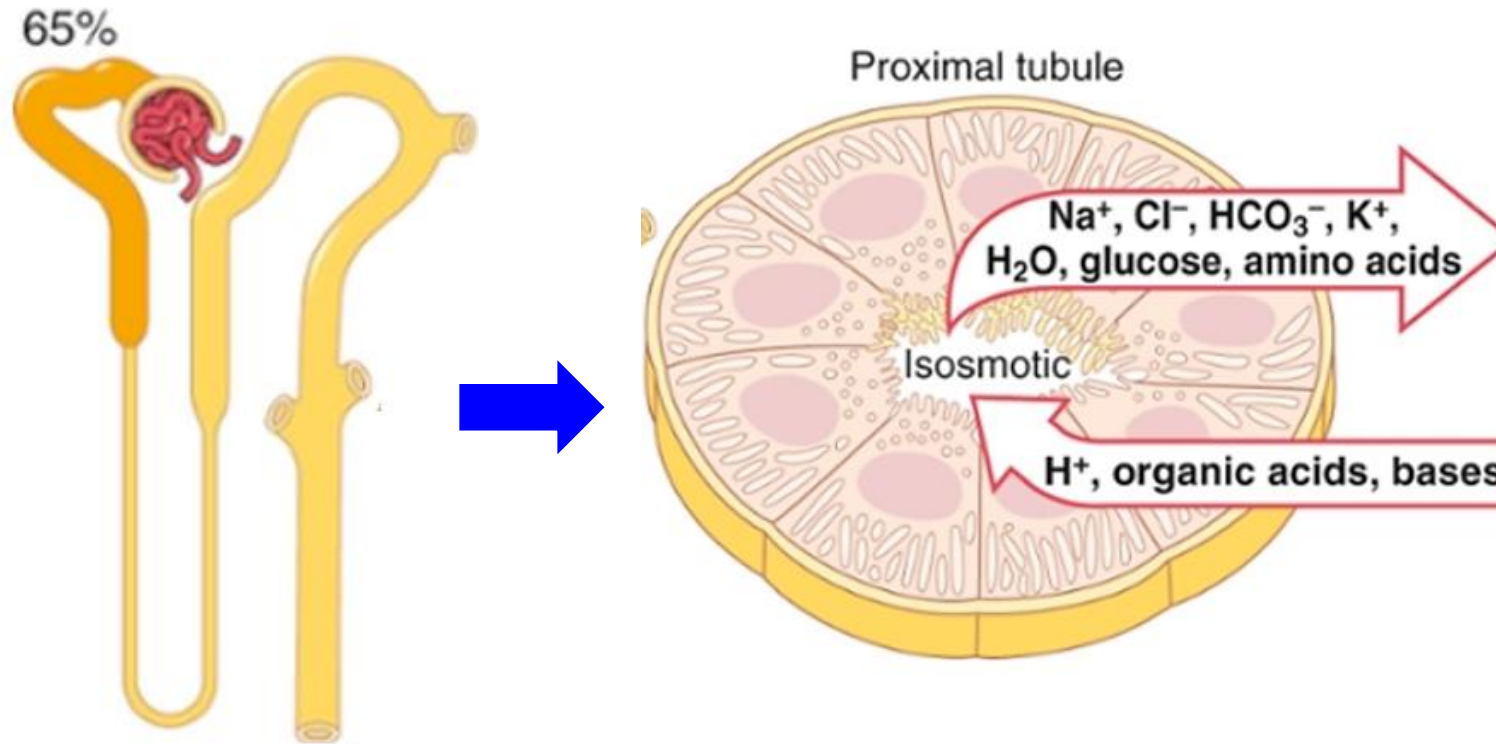
- As concentration of substance increases, the rate of simple diffusion increases.
- Once the transporters become saturated, the transport maximum(V_{max}) is attained in facilitated diffusion.

Primary vs Secondary Active Transport

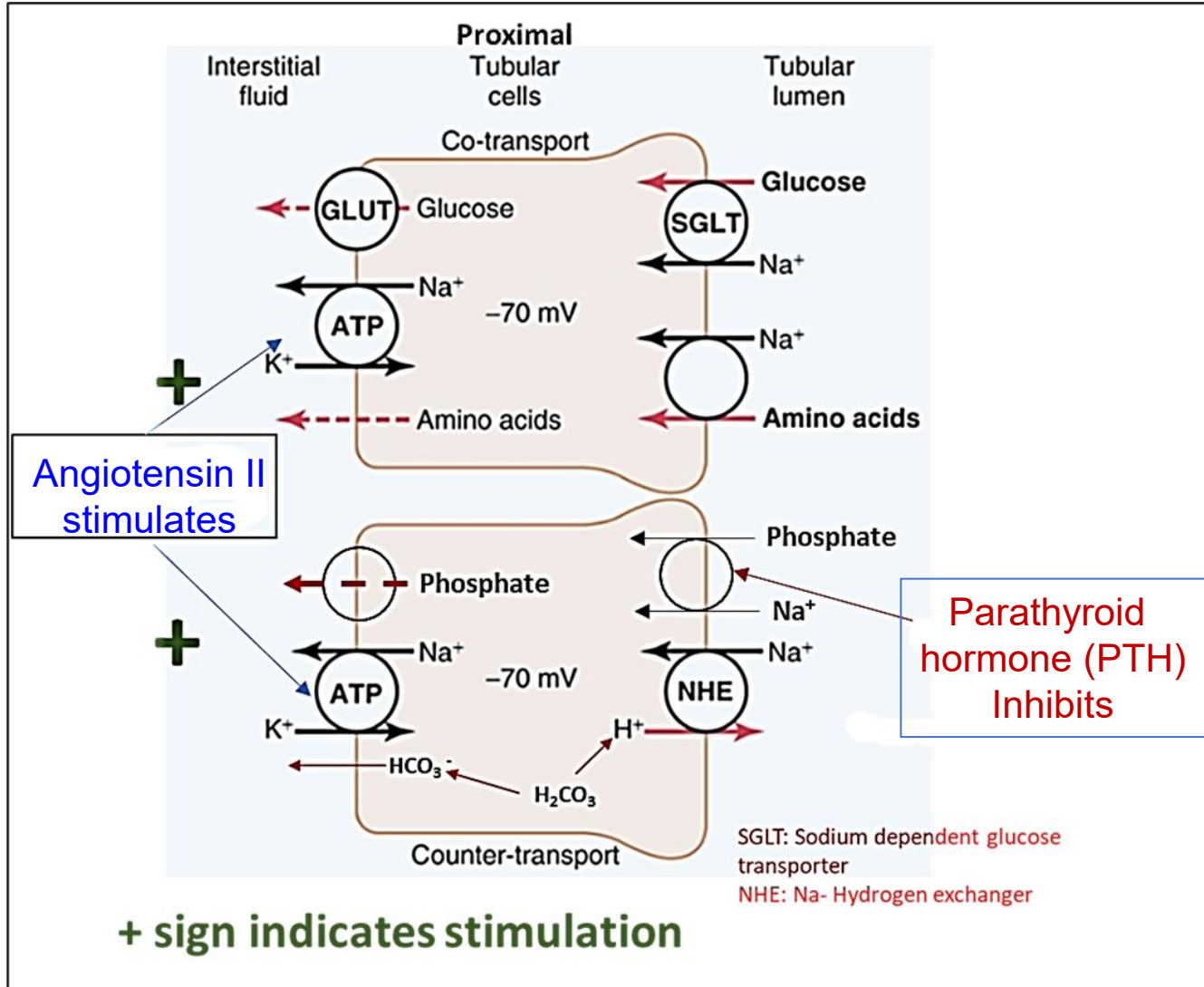
- Primary active transport: ATP is consumed directly by the transporting protein (e.g. Na/K-ATPase pump).
- Secondary active transport depends indirectly on ATP as a source of energy. E.g. the Na-glucose symporter. This process depends on ATP utilized by the Na/K-ATPase pump, creating a concentration gradient for sodium, allowing glucose to move up a concentration gradient.
- Transporters are primary sites of action for many drugs and hormones. Treatment of hypertension is associated with these drugs blocking the physiological actions of the transporters.

Tubular Mechanisms of Reabsorption and secretion in different segments of the nephron are explained in following discussion

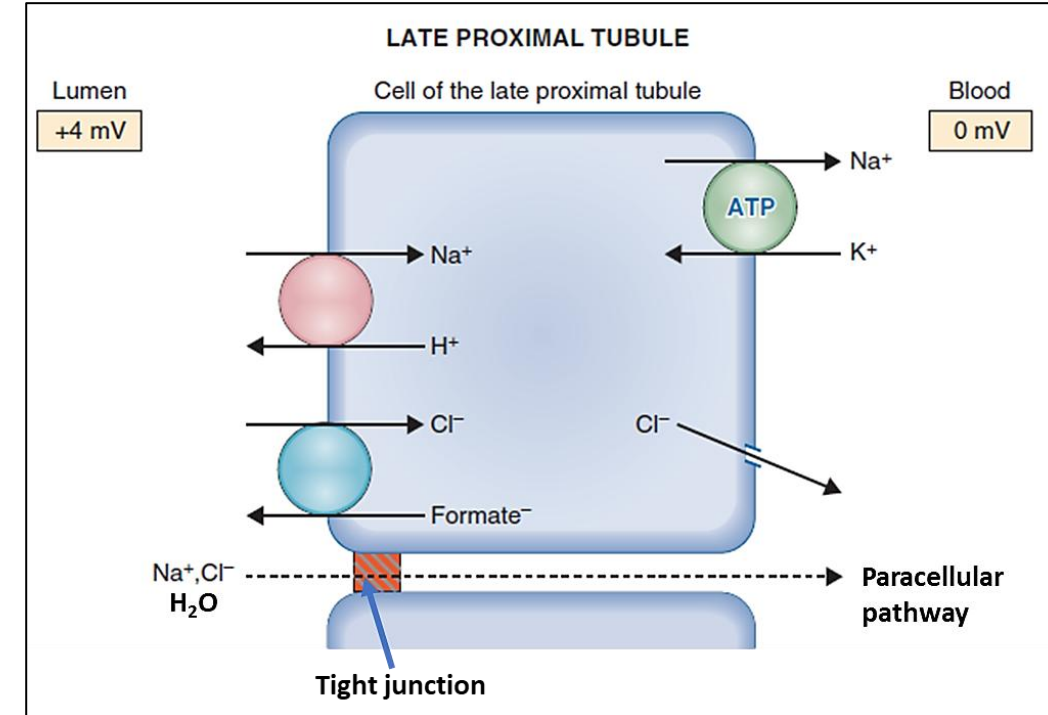
Reabsorption in Proximal Convoluted Tubules (PCT)



PCT: Sodium (Na^+) reabsorption



PCT: Water Reabsorption and some Na^+ & Cl^- reabsorption



Sodium (Na^+) reabsorption in PCT

- **“Filtered load”** is the amount of any substance that filters from the glomerulus and enters the Bowman space per unit time (mmol/min or mg/min).
- **Approximately 2/3 (approximate 67%)** of the filtered sodium is reabsorbed in the proximal tubule by various transporters without any hormonal influence.
- The basolateral Na/K-ATPase creates the gradient for sodium entry into the cell and its removal from cell back into the bloodstream.
- **Angiotensin II** stimulates the basolateral Na/K-ATPase and favors the fraction of sodium reabsorbed in the proximal tubule.
- **Parathyroid hormone (PTH)** inhibits Na-P transporter. Elevated PTH in parathyroid tumors reduces phosphate reabsorption and increase phosphate excretion in urine.

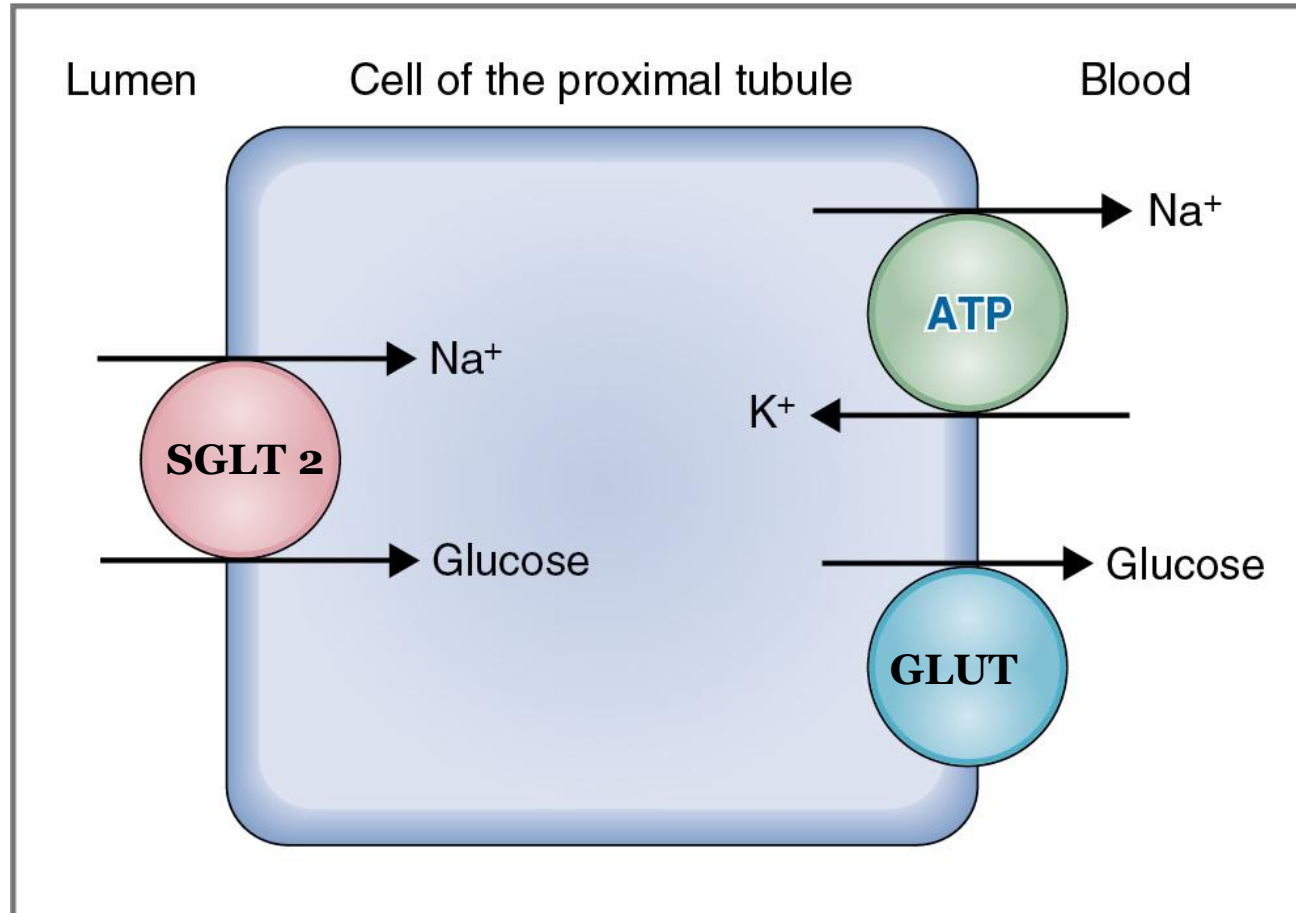
PCT-Water and Electrolytes Reabsorption

- Proximal tubules have a high permeability for water and **about 2/3 of the filtered water, potassium and chloride** follow the sodium.
- The **chloride concentration rises** slightly through the proximal tubule because of the large percentage of bicarbonate reabsorbed here.
- Osmotic flow of water in the proximal tubules occurs through *tight junctions* between the epithelial cells (**paracellular pathway**), and through the cells themselves(**transcellular pathway**).

PCT- Urea Reabsorption

- As water is reabsorbed from tubule, urea concentration in the tubular fluid increases
- This increase creates a concentration gradient favoring the passive reabsorption of urea paracellularly.
- Urea is formed from ammonia, as a nitrogenous waste. In plasma, it is called blood urea nitrogen (BUN). Its plasma concentration is useful indicator of renal function.

PCT: Glucose Reabsorption

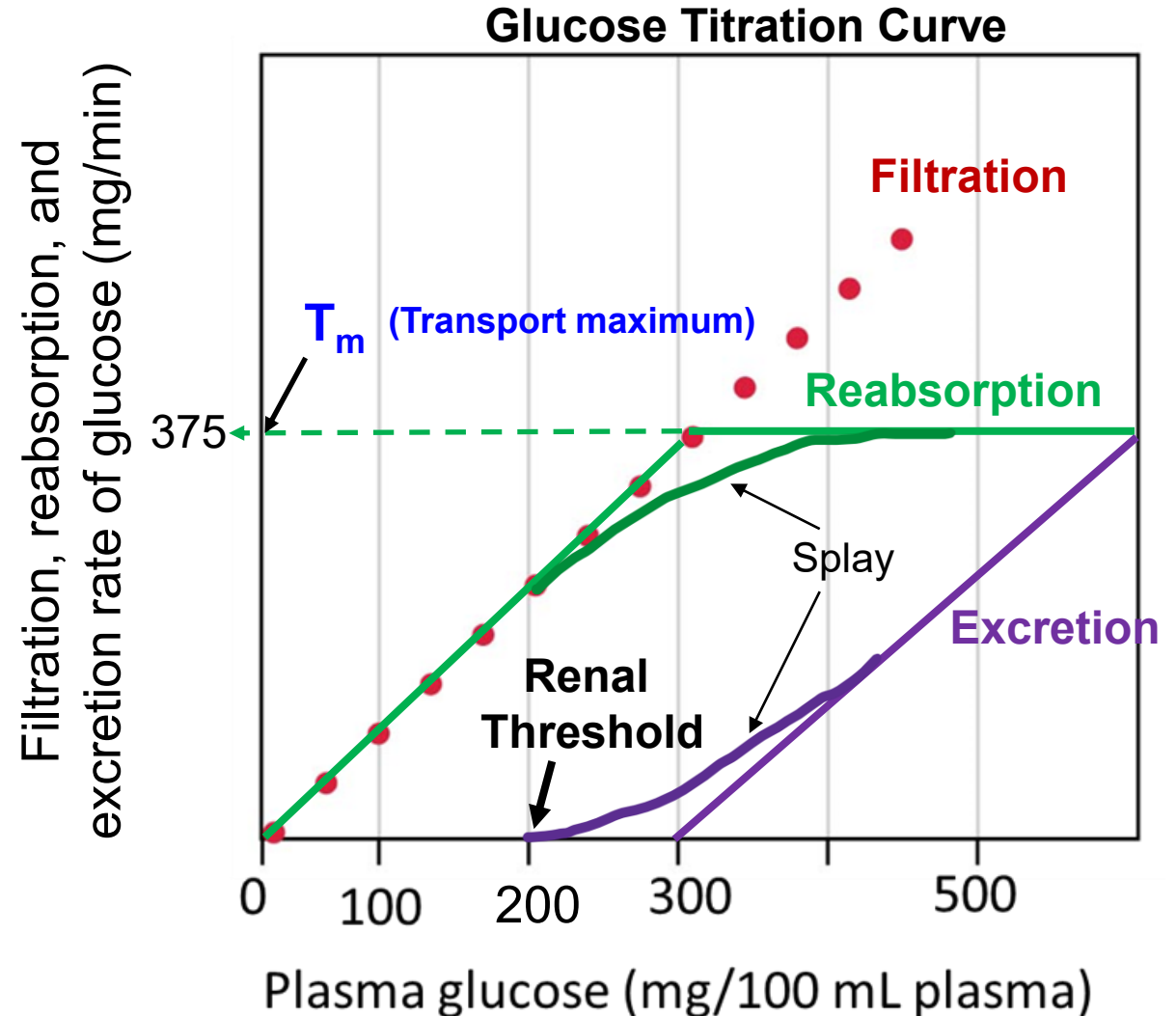


- Glucose is **completely** reabsorbed
- No hormones involved
- No glucose reabsorption occurs at other sites in the kidney.

SGLT = Na^+ -glucose cotransporter (secondary active transport)
GLUT = Glucose transporter (Facilitated diffusion)

Diabetes Mellitus: Saturation of Glucose Transporter

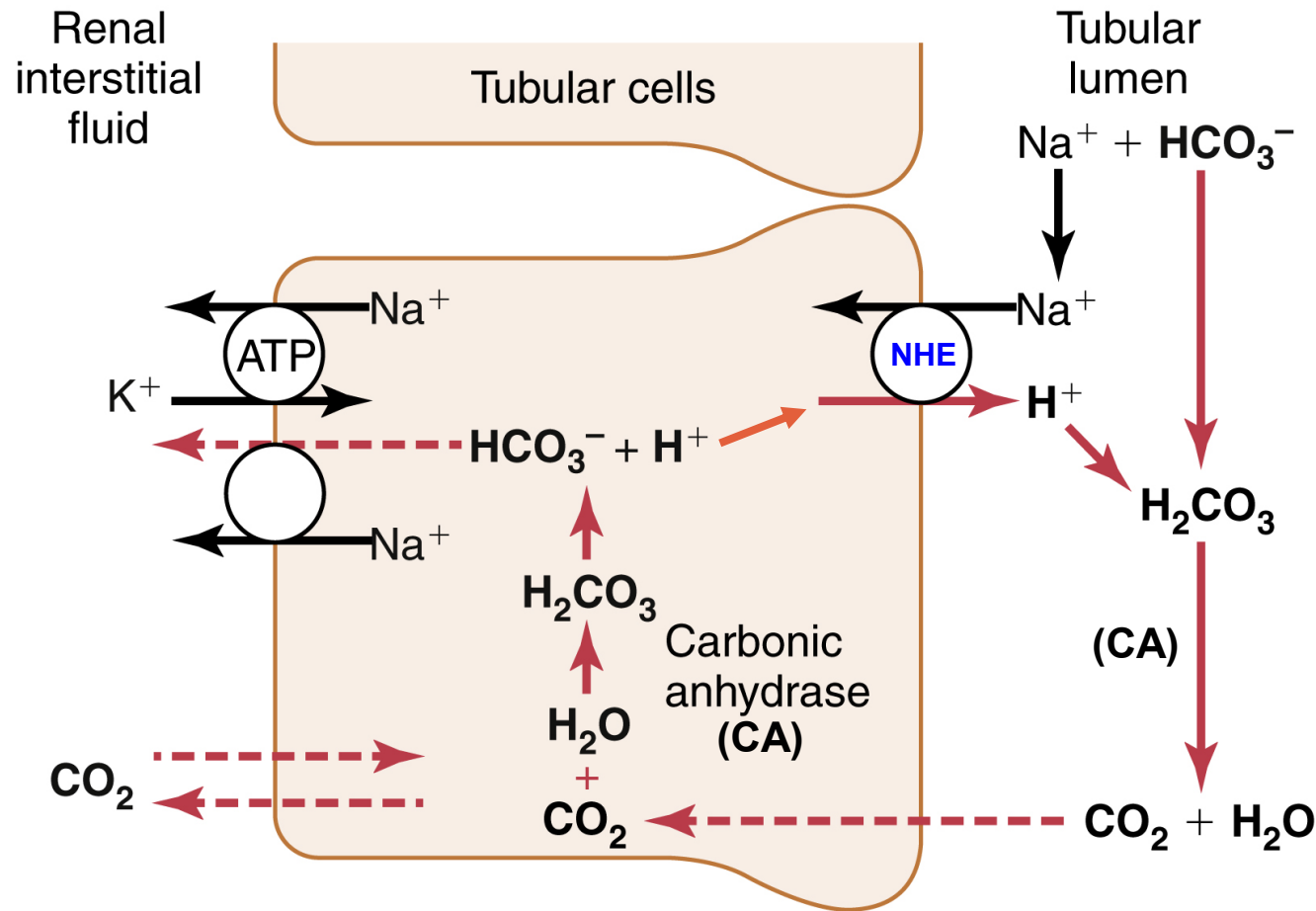
- Filtration of glucose is **proportional** to the plasma concentration. Filtration does not saturate.
- Reabsorption of glucose is proportional to plasma concentration until the **transport maximum (T_m)** is reached (saturate).
- Excretion = Filtration – Reabsorption. Glucose excretion is zero until the renal threshold is reached. Renal glucose threshold is the plasma glucose concentration beyond which glucose starts appearing in urine.
- The **unabsorbed glucose** inside the renal tubules holds water and causes osmotic diuresis, manifested as **polyuria** (increased urine volume)



Glucose Reabsorption in PCT

- Filtered glucose moves into the epithelial cells via the apical secondary active transport linked to sodium called Sodium-Glucose transporter (SGLT). SGLT2 subtype is involved in this process in the early PCT and the SGLT1 subtype in the late PCT. Glucose uptake by the epithelial cells generates a concentration gradient that drives facilitated diffusion via Glucose transporters (**GLUT2 and GLUT1**) transporters in the basolateral surface.
- Glucose is completely (100%) reabsorbed in the PCT at lower plasma glucose concentrations and until plasma glucose reaches the renal threshold of glucose. In diabetes mellitus, glucose appears in urine because the plasma glucose exceeds this renal threshold.
- Renal glucose threshold is 200 mg/dL (200 mg/dL of serum glucose concentration).
- Glucose reabsorption has saturation. Maximum amount of glucose reabsorbed is called Transport maximum (T_m) glucose. Normal value of T_m is 375 mg/min.
- Theoretically, if the amount of filtered glucose exceeds T_m glucose, glucose must appear in urine. However, in real life, glucose starts appearing in urine much before this value (glucose starts appearing in urine above renal threshold of glucose). This is shown as splay in the glucose titration curve.
- Reason for splay: Not all nephrons have the same glomerular size or the PCT length. Therefore, filtering and reabsorbing capacities of nephrons varies. The nephrons with relatively high filtration rates and low glucose absorptive rates excrete glucose before reaching T_m glucose.

PCT- Bicarbonate Reabsorption



Every bicarbonate converted into CO_2 in the tubular lumen is recovered back into blood. This is known as **conservation of filtered bicarbonate**.

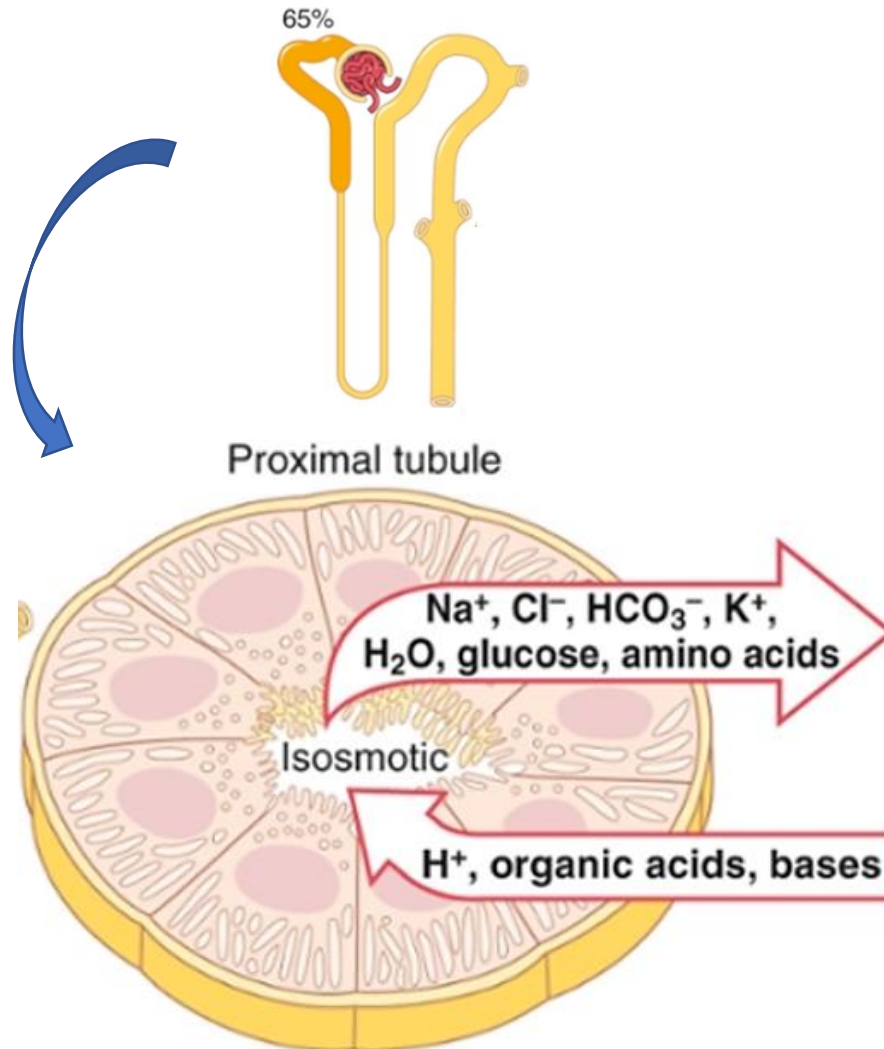
Whatever bicarbonate present in filtrate moves back to the blood. Hence there is **NO net gain or loss of bicarbonate** in the blood.

NHE=Sodium-Hydrogen Exchanger

PCT: Bicarbonate Reabsorption

- Excreting HCO_3^- causes the ECF to become acidic, so the body must recover 100% of the filtered HCO_3^- load. The PCT recovers ~80% of total.
- Because HCO_3^- is anionic, it cannot diffuse freely across membranes.
- The PCT secretes **equal H^+ for each HCO_3^-** . Carbonic anhydrase (CA) converts the H_2CO_3 to CO_2 and H_2O . Both molecules are then recovered by simple diffusion.
- The PCT can also actively secrete H^+ into the tubule using a H^+ pump (H^+ ATPase) in certain conditions to excrete acid.
- Acetazolamide, a carbonic anhydrase (CA) inhibitor, reduces bicarbonate reabsorption and the activity of the Na^+/H^+ antiport in the PCT. This prevents some amount of sodium to remain in the tubular lumen and hold water. This water is excreted in the urine. Therefore, this drug can be a diuretic, a drug that increases urine volume. Acetazolamide is used in the treatment or prophylaxis of altitude sickness where urinary bicarbonate excretion is helpful to offset acute respiratory alkalosis.

PCT: Reabsorption Summary



Major Substances Reabsorbed in PCT are:

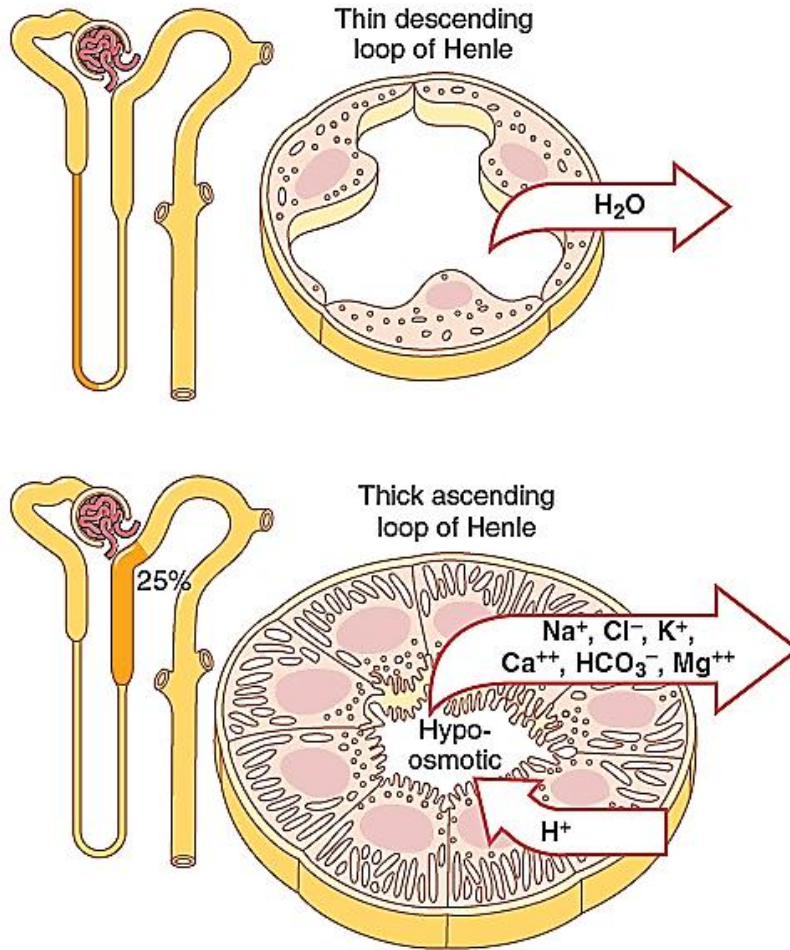
- Approximately **66% of filtered sodium, potassium, water**, and a slightly lower percentage of chloride.
- All filtered glucose.
- All filtered amino acids.
- Most of filtered bicarbonate.

Solutes reabsorbed at basolateral surface enter peritubular capillary blood by Starling's forces of fluid movement.

Proximal Tubular Reabsorption: Additional Features

- Fanconi syndrome: Clinical condition in which generalized reabsorption defect occurs in the PCT. Reduced solute reabsorption results in increased excretion of amino acids, glucose, HCO_3^- , PO_4^{3-} , and other substances in urine. This may lead to metabolic acidosis, hypophosphatemia, hypovolemia, growth retardation and osteopenia. The causes for this defect may be hereditary defects, nephrotoxins, ischemia, or drugs.
- **Glomerulotubular Balance:** The PCT recaptures about two-thirds of the filtered sodium under all conditions and this is referred to as glomerulotubular balance. Conserving this amount of sodium helps protect the extracellular volume despite any changes that may occur in GFR. The oncotic pressure of peritubular capillary (π_c) is the most important driving force for reabsorption in the PCT.

Reabsorption in Loop of Henle



Consists of three functionally distinct segments

Thin Descending Segment

- Permeable to water
- Impermeable to solutes

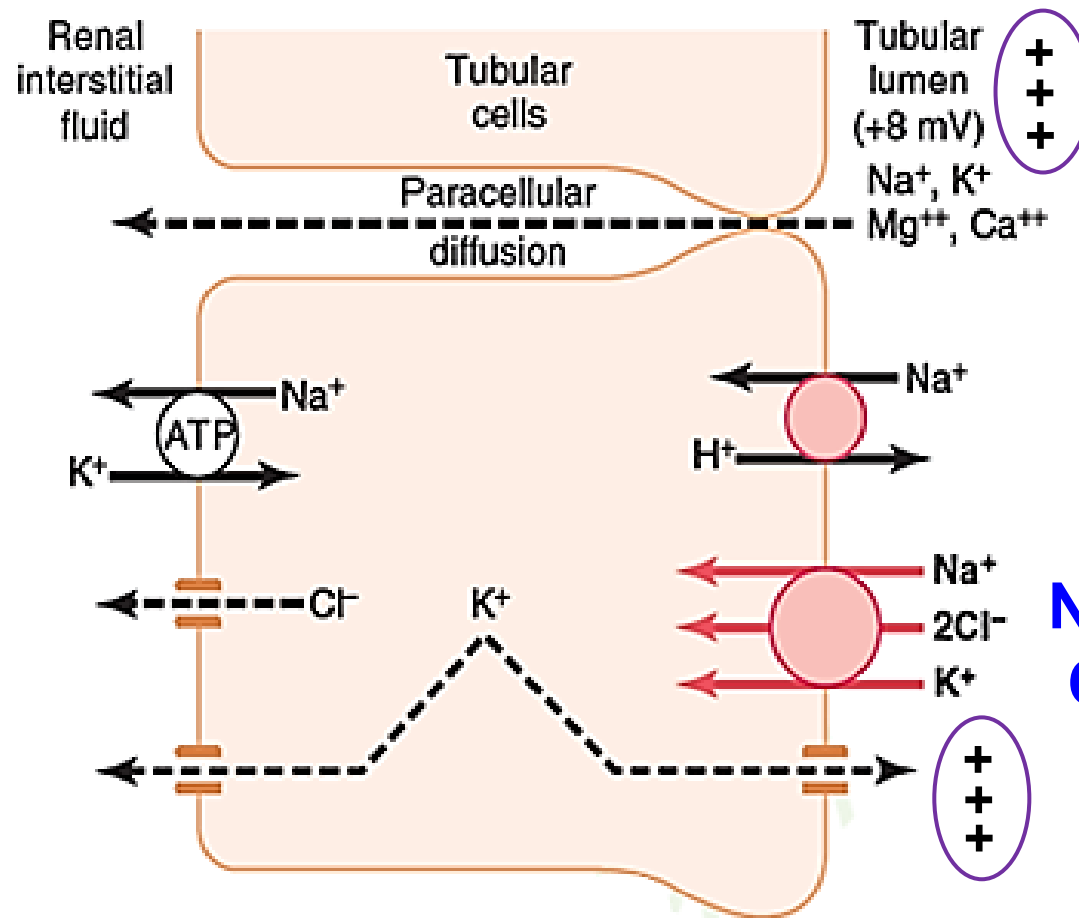
Thin Ascending Segment

- Impermeable to water

Thick Ascending Segment

- Impermeable to water
- Reabsorbs **25%** of the filtered load of sodium, potassium and chloride

Loop of Henle: Thick Ascending Segment Reabsorption



Loop diuretics like Furosemide block Na-K-2Cl cotransporters cause loss of sodium, potassium and calcium in urine.

Na-K-2Chloride Cotransporter

Bartter syndrome:

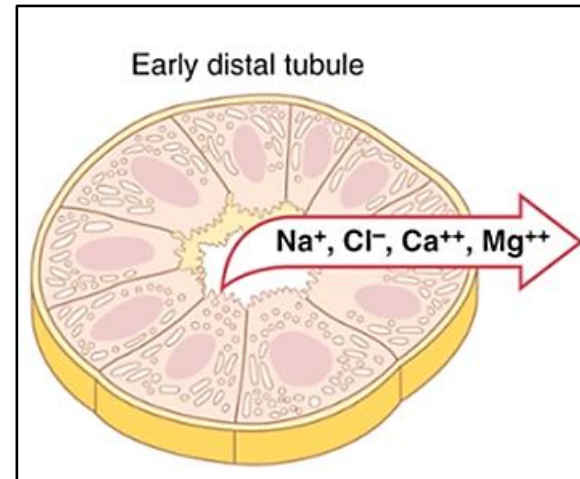
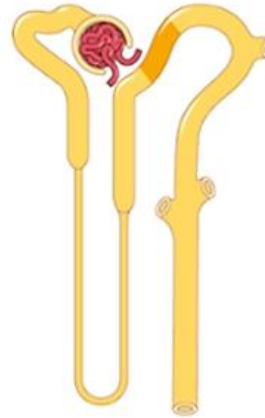
- Autosomal recessive disorder
- Reduced function of Na-K-2Cl
- Hypokalemia, hypocalcemia, calciuria & Metabolic alkalosis.

Thick Ascending Segment of Loop of Henle Reabsorption

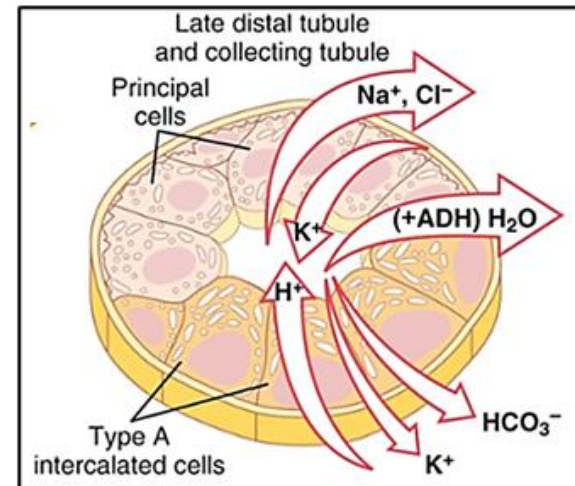
- Na/K-ATPase pump creates a low Na^+ in the epithelial cell that provides a favorable gradient for the movement of Na^+ across the luminal surface into the cell via **Na/Cl/K co-transporter**. About 25% of filtered sodium, potassium and chloride are reabsorbed by this mechanism.
- The low intracellular Na^+ concentration, also drives the Na/H^+ antiport causing H^+ secretion and bicarbonate absorption.
- K^+ channel of the luminal membrane causes diffusion of K^+ back into lumen which creates a positive luminal potential. This positivity of the lumen promotes Na^+ , Ca^{2+} , and Mg^{2+} reabsorption via **paracellular pathway**.
- Loop diuretics block Na-K-2Cl co-transporter and increase urinary loss of sodium, chloride, potassium and water. Inhibiting this cotransporter also decreases the calcium reabsorption into the blood and leads to **hypocalcemia**. Loop diuretics are used to remove large volume of plasma rapidly and relieve symptoms in heart failure patients.
- Antidiuretic hormone (ADH) stimulates Na-K-2Cl cotransporters.

Reabsorption in the Distal Tubule and Collecting duct

Approximately 10%
filtered sodium is
reabsorbed in these
regions



Early distal tubule
has Na-Cl symporter

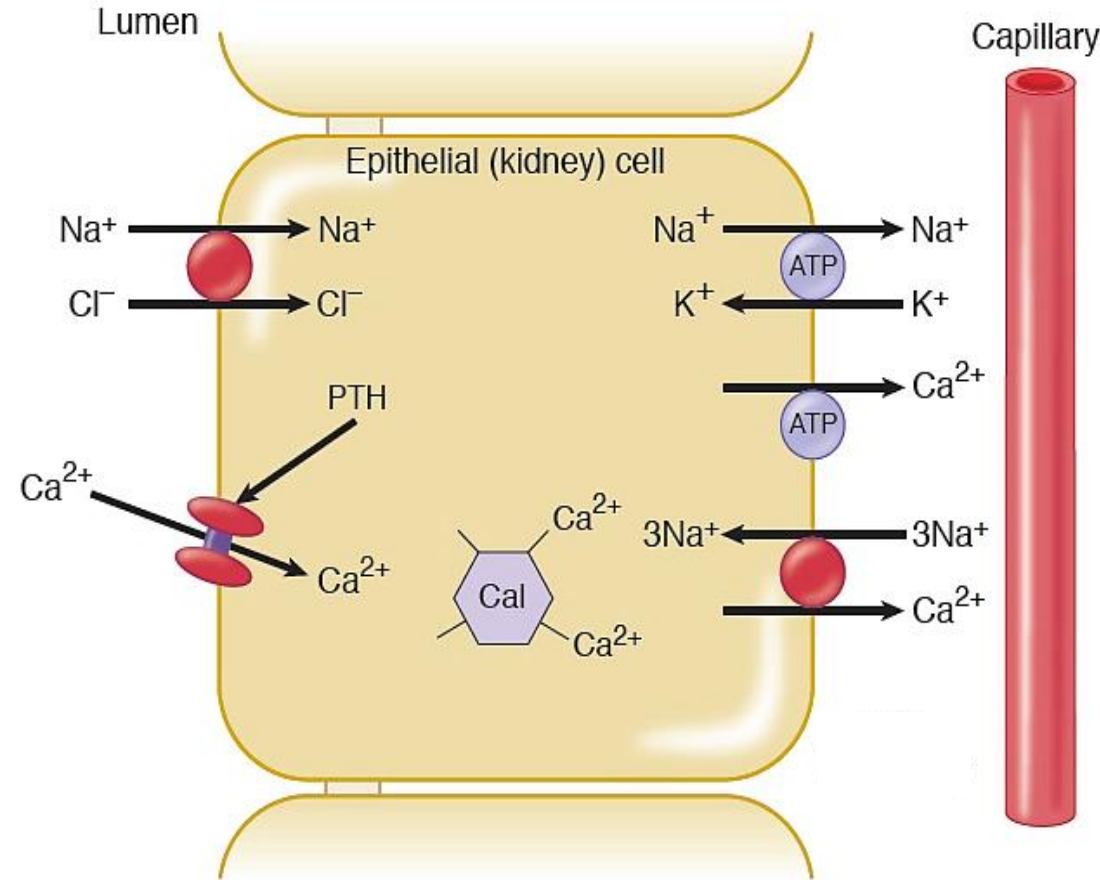


Late distal tubule
and collecting duct
have similar
mechanisms of
transport

Reabsorption in Early Distal Tubule

Sodium-chloride symporter

Thiazide Diuretics block this symporter.
Preventing Na^+ influx makes inside of the cell more negative and favors calcium reabsorption and cause **hypercalcemia**.



Reduced functioning of Na-Cl symporter occurs in Gitelman syndrome

Parathyroid hormone (PTH) stimulates the calcium channels and increases calcium reabsorption from DCT and produces hypercalcemia.

Early Distal Tubule Reabsorption

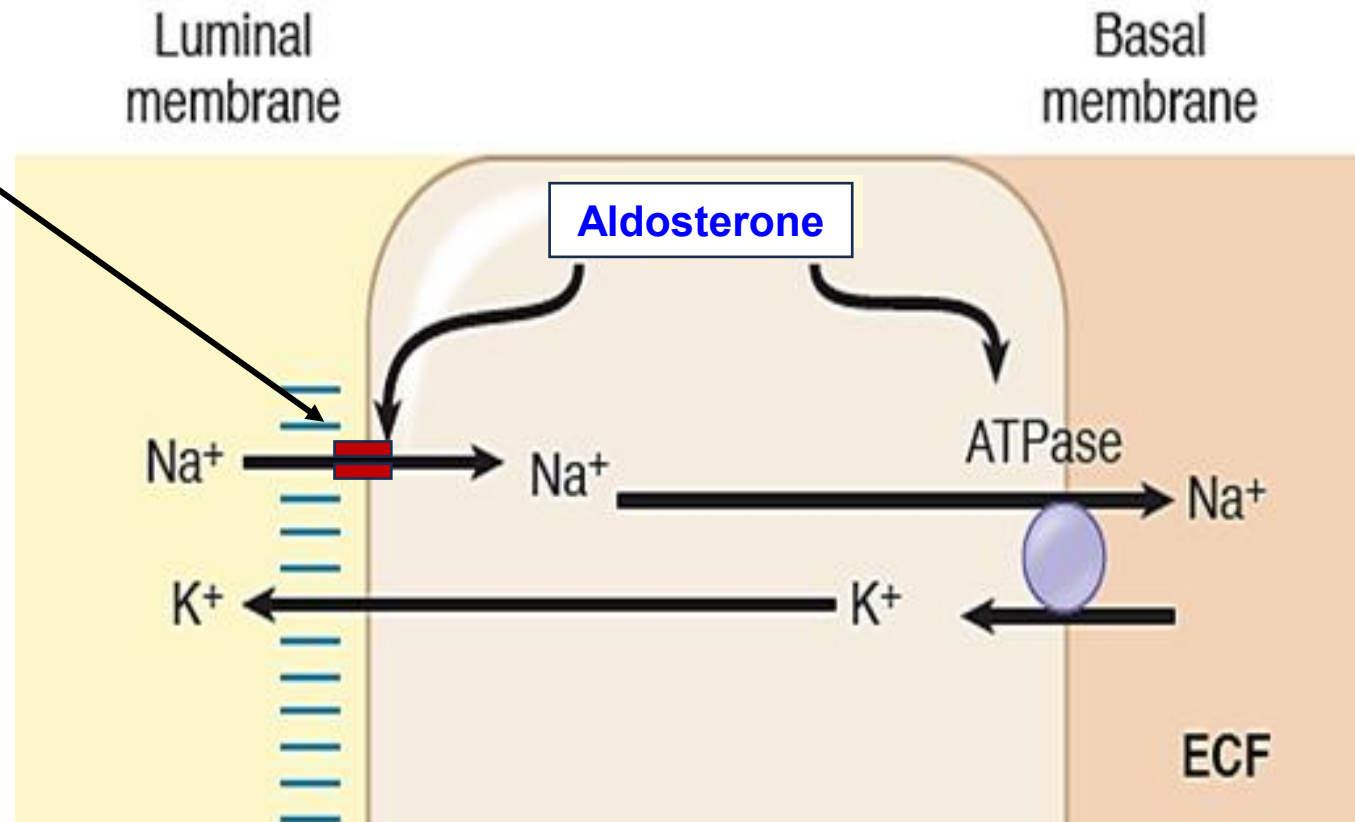
- Sodium-chloride symporter in the luminal surface is the major mechanism of sodium reabsorption in this part of nephron.
- Luminal Ca^{2+} is also reabsorbed calcium channels which are stimulated by the parathyroid hormone (PTH).
- The **thiazide diuretics** inhibit Na-Cl symporter in the distal tubule and are widely used as diuretics to treat hypertension and heart failure.
- Reason for thiazide diuretics causing hypercalcemia: By inhibiting Na^+ influx into the cells, thiazide diuretics produce hyperpolarization of the epithelial cells. This hyperpolarization makes the electrochemical driving for Ca^{2+} movement from the lumen into blood via the epithelial cells. This reabsorption of calcium results in **hypercalcemia**. Note the difference with loop diuretics which cause hypocalcemia.
- Gitelman syndrome: Condition due to reduced functioning of Na-Cl cotransporters. An autosomal recessive disorder. These patients will have hypokalemia, hypocalciuria (because of increased Ca^{2+} reabsorption in the distal nephron) and metabolic alkalosis.

Reabsorption in Late Distal Tubule & Collecting Duct Principal cells

ENaC (Epithelial Sodium Channel)

Principal Cells Reabsorb Sodium and Secrete Potassium

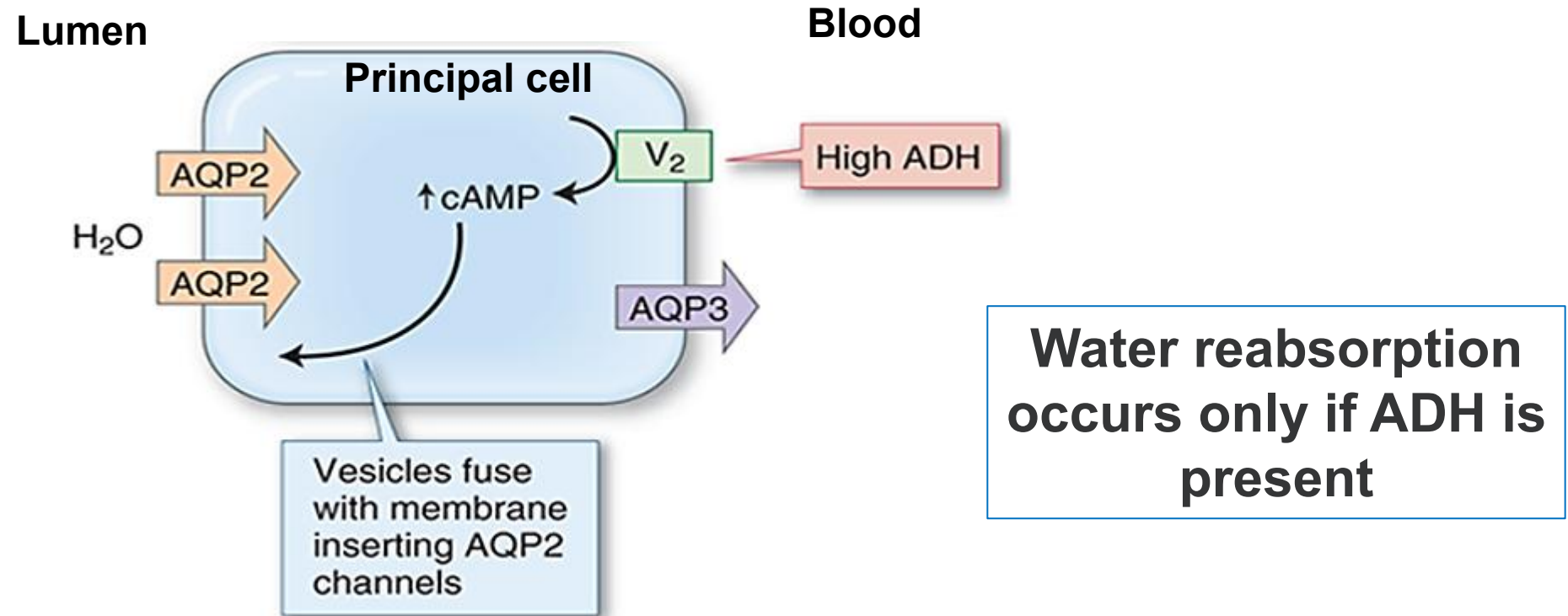
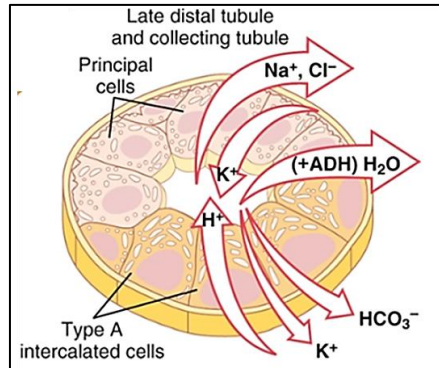
Sodium reabsorption here is facilitated by **aldosterone**; blocked by potassium-sparing diuretics



Liddle syndrome: Mutation increases activity of ENaC resulting in increased reabsorption of Na⁺ and increased secretion of K⁺.

Reabsorption in Late Distal Tubule & Collecting Duct Principal cells

Water reabsorption is facilitated by anti-diuretic hormone (ADH)



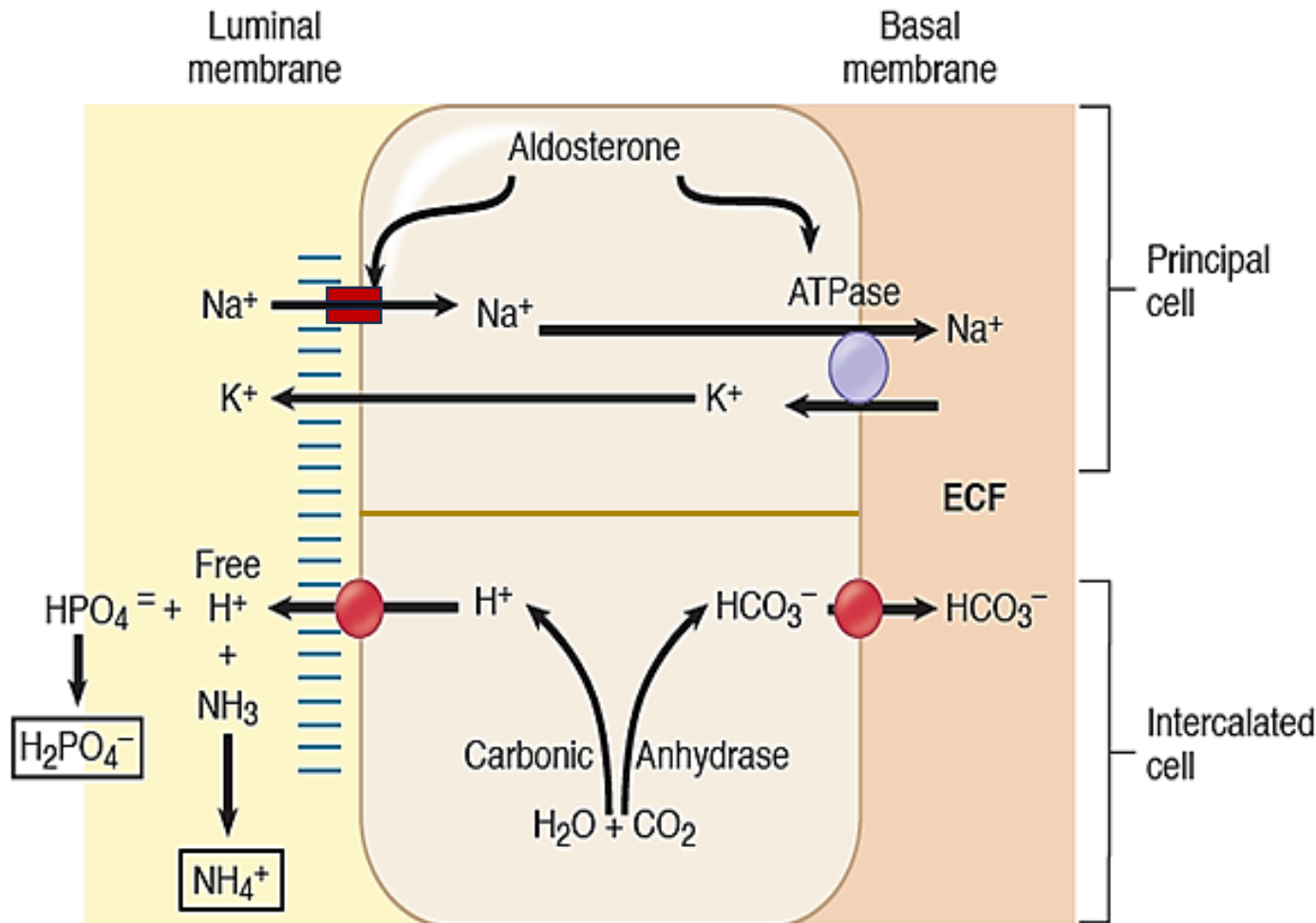
Sodium reabsorption in Principal cells by aldosterone

- Luminal membrane contains sodium channels (ENaC) → influx of sodium down its concentration gradient (created by Na/K⁺-ATPase)
- Some chloride does not follow sodium, creating a negative luminal potential causing potassium secretion
- Aldosterone activates the mineralocorticoid receptor on these cells, having the following effects:
 1. Increases ENaC channel activity
 2. Stimulates Na/K⁺-ATPase
- Excess aldosterone (hyperaldosteronism) causes hyponatremia, hypokalemia and metabolic alkalosis.
- Potassium-sparing diuretics block ENaC and thus elevate serum potassium.

Water reabsorption in Principal cells is by ADH

- Principal cells express water channels called aquaporins, which are regulated by anti-diuretic hormone (ADH).
- ADH acts on V2 receptors to cause insertion of aquaporins which in turn causes water reabsorption.
- Absence of ADH results in excretion of large volume of urine.

Late Distal Tubule & Collecting Duct Intercalated cells: Hydrogen ion secretion and Bicarbonate reabsorption



- Intercalated cells secrete H^+ and cause reabsorption of a **NEW** Bicarbonate ion.
- This bicarbonate is extra to filtered bicarbonate.
- This mechanism helps to secrete acid and **replenish** bicarbonate buffer when needed.

Aldosterone also stimulates H^+ -ATPase

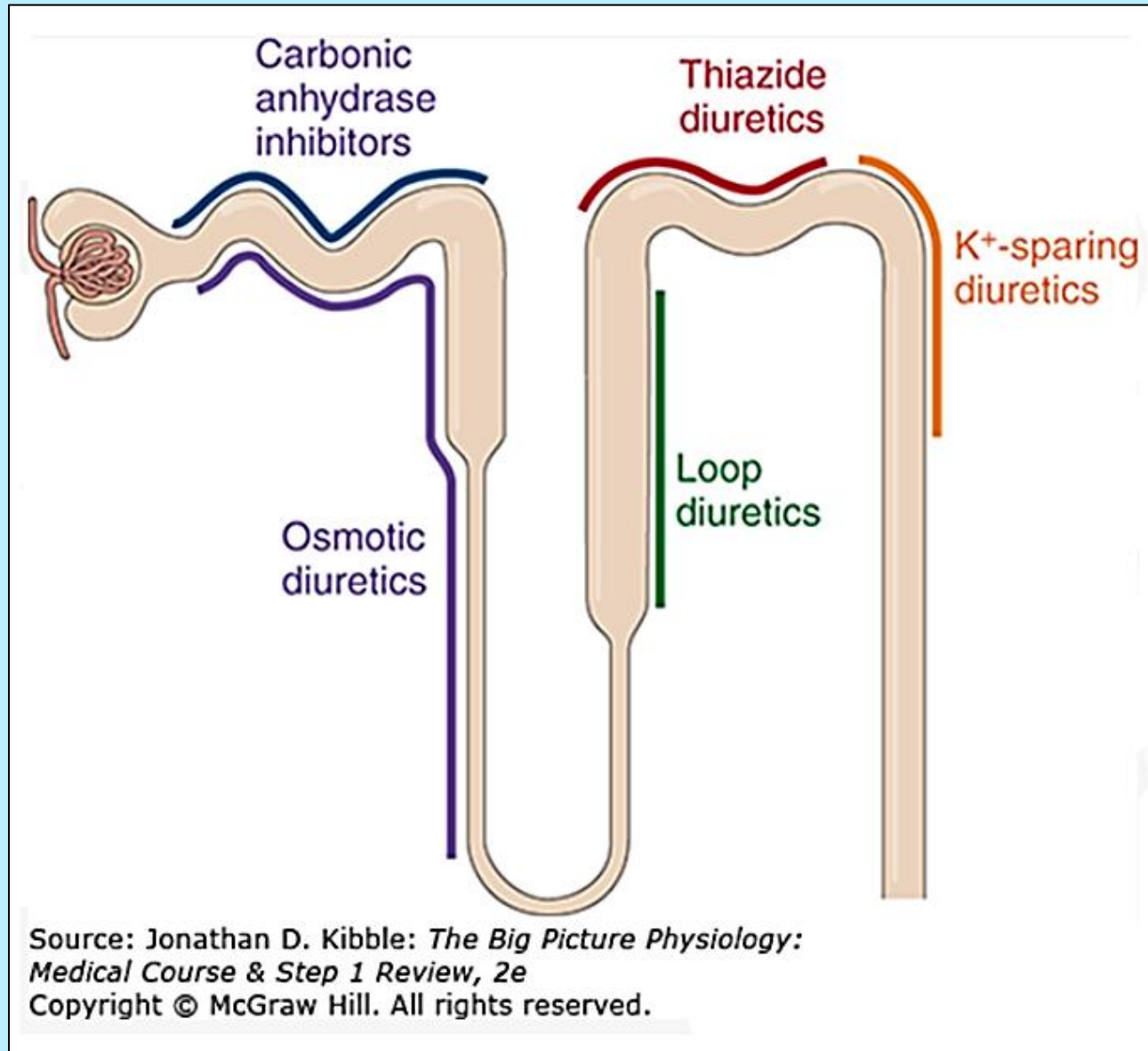
Late Distal Tubule & Collecting Duct- Intercalated cells

- Luminal membrane contains a H^+ -ATPase, which pumps H^+ into the lumen. Most of the H^+ is eliminated from the body via buffers, phosphate and ammonia.
- H^+ pumped into the lumen binds to phosphate to form deprotonated phosphate, which helps in eliminating H^+ from the body.
- H^+ can also combine with ammonia to form ammonium ions, which are poorly reabsorbed and is thus excreted.
- For every H^+ secreted and buffered by above mechanisms, one NEW bicarbonate is added to the body (new bicarbonate).
- Aldosterone stimulates H^+ -ATPase of intercalated cells. Thus, excess aldosterone causes excess H^+ secretion and Metabolic Alkalosis in addition to hypokalemia.

Effects of Hormones on solute reabsorption

Hormone	Site of action	Effects
Aldosterone	Late DCT and Collecting duct	↑Sodium reabsorption; ↑Potassium secretion & ↑H ⁺ secretion
Angiotensin II	PCT	↑Sodium reabsorption
Atrial Natriuretic Peptide (ANP)	Collecting duct	↓Sodium reabsorption
Antidiuretic Hormone (ADH)	Late DCT and Collecting duct	↑Water reabsorption; ↑urea reabsorption (only in inner medullary collecting duct)
Parathyroid hormone (PTH)	PCT and DCT	In PCT: ↓Phosphate reabsorption; In DCT: ↑Calcium reabsorption

Site of action of major groups of diuretics



Note: Osmotic diuretics (like mannitol) hold and prevent water reabsorption in the proximal renal tubule and induce diuresis.

Key Concepts

- PCT reabsorbs about 2/3rd of filtered Na^+ , K^+ , and water and nearly all of the filtered glucose and amino acids. It also secretes many organic anions and organic cations.
- The transport of water and most solutes across tubular epithelia is dependent on active reabsorption of Na^+ by Na-K-ATPase in the basolateral surface.
- The thick ascending limb is a water-impermeable segment that reabsorbs Na^+ via a Na–K–2Cl cotransporter in the apical cell membrane and Na^+/K^+ -ATPase in the basolateral cell membrane.
- The distal convoluted tubule epithelium is water impermeable and reabsorbs Na^+ via a thiazide-sensitive apical membrane Na–Cl cotransporter.
- Principal cells in the late DCT and cortical duct reabsorb Na^+ and secrete K^+ through epithelial sodium channels (ENaC) which is stimulated by aldosterone.
- Intercalated cells in collecting tubules secrete H^+ ions and add NH_4^+ bicarbonate to the blood.
- Renal tubules are the pharmacological site of action for diuretics.

DIY Task: Please read the case and answer the questions.

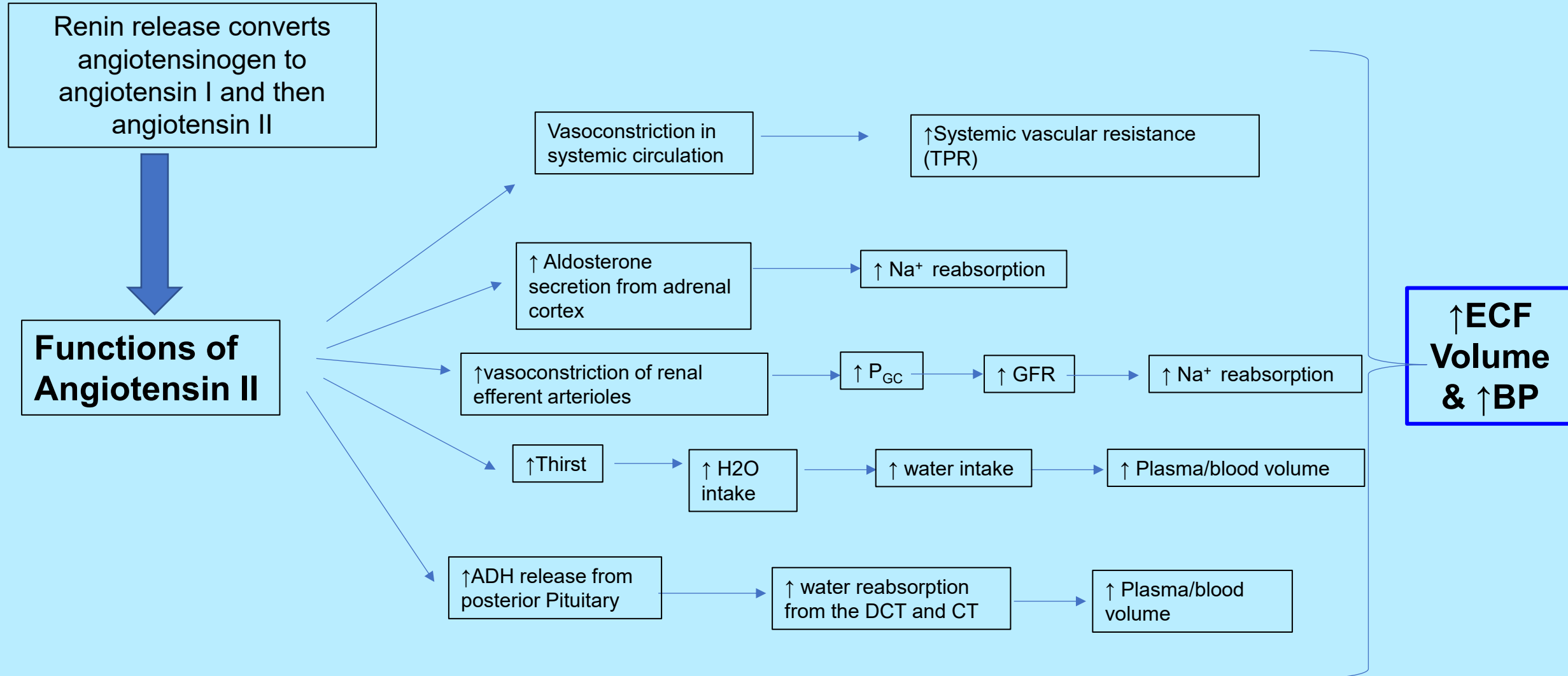
A 14-year-old girl comes to a specialist with an initial diagnosis of primary hyperaldosteronism. Her blood pressure is 160/120 mm Hg and pulse is 84/min. She has hypokalemia (serum K⁺ of 2.6 mEq/L), and alkalosis. Her urine report shows significant amount of K⁺ excretion despite her hypokalemia. Her plasma renin is lower than normal. Her aldosterone secretion rate was found to be abnormally low, even when on a low-sodium diet. She has a family history of genetic disease effecting both her parents and siblings. Eventually, she develops renal failure after 30 years later. She receives a kidney transplant. The transplant lowered her blood pressure and corrected her hypokalemia.

1. Why the initial diagnosis is of hyperaldosteronism in the patient?
2. What is an unusual observation related to hyperaldosteronism?
3. Why does a patient with hyperaldosteronism has hypertension, hypokalemia, and metabolic alkalosis?
4. What is the genetic disorder in the patient?
5. What class of diuretics would be helpful in correcting the clinical condition before transplantation?
6. What strongly indicates that the patient's problem was an intrinsic renal defect?

DIY Task: Answers

1. Why the initial diagnosis is of hyperaldosteronism in the patient? Ans: Because this patient has hypertension, hypokalemia, and metabolic alkalosis with a decrease in plasma renin. These parameters strongly suggest primary hyperaldosteronism.
2. What is an unusual observation related to hyperaldosteronism? Ans: The aldosterone secretion rate was abnormally low.
3. Why does a patient with hyperaldosteronism has hypertension, hypokalemia, and metabolic alkalosis? Ans: Aldosterone causes increase sodium (as well as water) reabsorption and K^+ secretion from the DCT principal cell, leading to hypernatremia(increased plasma sodium) and hypertension. It also causes H^+ secretion from the intercalated cell, leading to metabolic alkalosis.
4. What is the genetic disorder in the patient? Ans: The patient has Liddle syndrome.
5. What class of diuretics would be helpful in correcting the clinical condition before transplantation? Ans: potassium-sparing diuretic drugs which block ENaC, are useful in correcting the clinical condition before transplantation.
6. What strongly indicates that the patient's problem was an intrinsic renal defect? Ans: The hypertension was improved, and the patient had a normal plasma $[K^+]$ after she received a new kidney.

Flowchart to show the physiological effects of Angiotensin II and its effect on ECF volume and blood pressure



Any Questions ?

How to reach me: Either walk into my office or email.

Subramanya Upadhyaya, MD.

Professor,

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